# THE FAMOUS STRATEGY GAME

# GAME INSTRUCTIONS

## BRIEF IDEA OF THE GAME

Welcome to the wicked world of Madballs – a physics-based puzzle game full of chaos, chain reactions, and evil-eyed objects!

In Madballs, you're the mastermind behind hilariously unpredictable machines. Your toolbox? A mischievous mix of grinning tennis balls, smug beach balls, heavy bowling balls, wobbly platforms, sneaky dominoes, shady crates, leaking buckets, and suspiciously cheerful balloons – all with faces that look like they're up to no good.
Your goal: combine these quirky parts to solve a series of increasingly challenging physics puzzles. Build chain reactions, trigger mechanisms, and make the impossible work – one level at a time.

**Game Modes**:

- **Puzzle Mode**: Progress through a series of handcrafted levels, each with a unique challenge. Use logic, timing, and a bit of madness to find creative solutions and move on to the next contraption.
- **Sandbox Mode**: Let your imagination run wild! Build your own machines and experiments using a powerful visual editor inspired by node-based systems like Graphviz. There are no objectives here – just freedom, fun, and glorious chaos.

Whether you're solving clever puzzles or creating the weirdest machine you can think of – Madballs is all about playful physics, surprising reactions, and wicked little smiles.

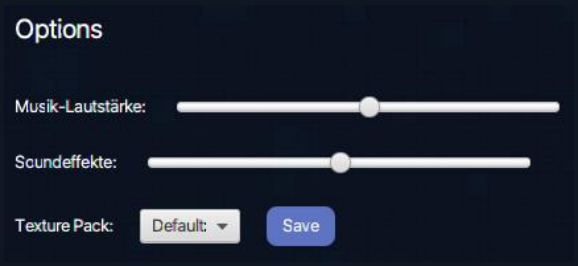
# GAME REQUIREMENTS

- Operating System: Linux
    - Partial support of MacOS & Windows10/11
- Monitor Resolution: At least 1920x1080

# THE TITLE SCREEN

From the Title Screen, you can navigate between the different game modes and access the Options Menu.
You can also return to the title screen at any time during gameplay (see the *Control Center Overview* section for details).

**Options-Menu**:




*Note: In the current version of the game, sound is not implemented. The sound settings are therefore purely visual and have no effect.*

**Texture Packs**

In the **Texture Pack** section, you can switch between the available visual styles:

- **Default Texture Pack**
- **Legacy Texture Pack**

To select a texture pack, simply choose one from the **dropdown menu** and confirm your choice by clicking the **Save** button, as shown in the image.

# CONTROLL CENTER OVERVIEW

You control the entire game using a small, centered button hub, as shown in the image above.
This interface changes depending on the current game mode. The example shown here is the Sandbox Mode Control Center, displaying all actions available in that mode.

Here's what each Button does:

- ▶ Start Simulation
  Begins the simulation based on your current setup.
- ■ Stop Simulation
  Ends the simulation and returns you to edit mode.
- ⚙ or ESC Options / Quick Menu
  Opens the in-game options where you can quit the game or return to the title screen.
- 🗑 / ⟳ Clear or Reset Level
  In Sandbox mode: 🗑 removes all placed objects.
  In Puzzle mode: ⟳ resets the current level to its original state.
- 🗁 Load Level (JSON)
  Loads a previously saved level file from your computer.
  Note: Loading levels is only available in Sandbox Mode.
- 💾 Save Level (JSON)
  Saves your current setup as a level file.
  This can also be done after completing a puzzle to share your solution with others.
- ↶ / ↷ Undo / Redo
  Lets you undo or redo recent actions such as object placement or movement.

# INVENTORY AND OBJECTS

The **Inventory** is located in the **right sidebar** and remains the same across all game modes.
All available *Mad Objects* are listed vertically within this panel.

Once the number of available objects exceeds the visible space, you can scroll through the inventory using the **mouse wheel** or a **trackpad**.
To the right of each object, you'll see a number indicating how many instances of that object type you can still place. If the count reaches **zero**, you can no longer use that type.

To place an object into the simulation area, simply **drag and drop** it from the inventory onto the workspace.

### Object Types

*Madballs* features a variety of object types, including:

- Tennis Ball
- Bowling Ball
- Beach Ball
- Platform
- Log
- Bucket
- Domino
- Crate
- **No-Build Zone** (red)
- **Win Zone**
- Balloon
- Win Platform

Each **ball** type has its own size, weight, and behavior, making them useful in different situations.

The **Log**, all **Zones**, **Platforms**, **Dominos**, and **Bucket** are **static objects** – they don't move unless acted upon by other objects.

The **Balloon** is a special object: it rises upward if there's no object above it, and will **pop** if it reaches a certain height.

🚫 **No-Build Zone (Red)**

As the name suggests, you cannot place objects **inside** this zone.
However, objects can **fall, roll, or fly** through it during simulation.

🏁 **Win Conditions**

When a designated **Win Object** touches the **Win Zone/Win Platform**, the level is completed successfully. By default, the **Beach Ball** acts as a Win Object, but this can be changed via the **Text Editor** (see section on Text Editor for more info).

## SIMULATION SPACE

The **Simulation Area** is where the core gameplay takes place.
Here, you can **place and observe objects** as they interact through our innovative physics system.

As mentioned earlier, you can add objects to this area using simple **drag & drop** from the inventory.

**Object Interaction**:

Once an object is placed, you have several ways to interact with it:

- **Move Objects**:
  If you want to reposition an object, just drag and drop it to a new location.
- **Rotate Objects**:
  To rotate an object, simply hover your mouse over it and use the **mouse wheel** or **trackpad gesture** to rotate it in place.

This gives you full control over how each object behaves and allows for precise puzzle-solving and creative building.

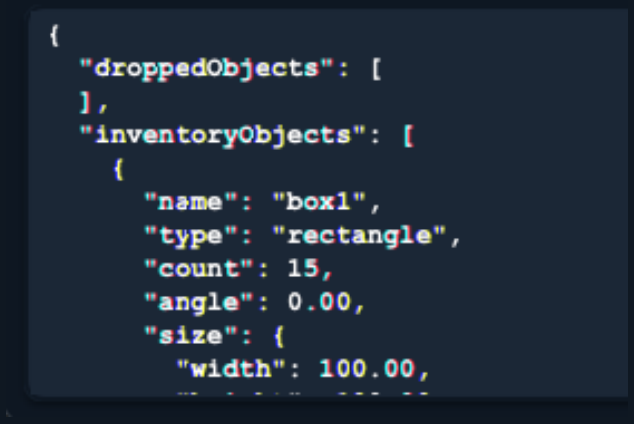
## SANDBOX MODE

In **Sandbox Mode**, you have complete creative freedom.
Design your own puzzles and share them with friends using the **Export** function (see *Control Center Overview* for details).

You can also **load puzzles** created by friends, your own custom challenges, or saved solutions to existing levels – loading is exclusive to this mode (see *Control Center Overview*).

At the bottom of the screen, you'll find our built-in **Text Editor**, which lets you directly modify the level's data. You can customize objects, turn them into **Win Objects**, or change their **physical properties**.

To learn how to use the syntax properly, refer to the **next chapter** for our Leveldata-Syntax wiki.




## LEVELDATA-SYNTAX WIKI

This page explains how levels are structured in Madballs using JSON. Each level file contains two main sections:

- "levelObjects" – pre-placed elements in a puzzle level
- "inventoryObjects" – available objects in Sandbox mode or puzzles

### Object Structure

Every object shares a common structure. Here's a breakdown:

```
{
  "name": "platform",
  "type": "rectangle",
  "position": { "x": 500.0, "y": 800.0 },
  "angle": 0.0,
  "size": {
    "width": 300.0,
    "height": 20.0,
    "radius": 0.0
  },
  "sprite": "madPlatform.png",
  "colour": "BLACK",
  "physics": {
    "density": 1.0,
    "friction": 0.4,
    "restitution": 0.1,
    "shape": "STATIC"
  },
  "winning": false,
  "count": 15
}
```

### 🔑 Field Reference

| Key | Description |
|---|---|
| name | The internal name of the object. |
| type | Shape type: "rectangle", "circle", or "bucket" (for custom shapes). |
| position | Object's coordinates on the simulation area. Not used in inventoryObjects. |
| angle | Rotation of the object in degrees (float). |
| size | Object dimensions. For circles, use radius; for rectangles/buckets, use width and height. |
| sprite | Optional path to the sprite file (e.g., "madBall.png"). Can be null. |
| colour | Visual color. Can be a color name or hex code (e.g., "#800080"). |
| physics | Defines the physical behavior of the object (see next section). |
| winning | Boolean. If true, object will trigger a win condition upon entering a winZone or touching a winPlatform. |
| count | Only in inventoryObjects: number of available instances. |

---

### ⚙️ Physics Properties

Under the physics key:

| Property | Description |
|---|---|
| density | Affects object mass. Heavier objects have higher density. |
| friction | Affects how the object slides. High values reduce movement. |
| restitution | Bounciness. 0.0 = no bounce, 1.0 = full bounce. |
| shape | Either "STATIC" (does not move) or "DYNAMIC" (moves & reacts to physics). |

---

### Object Types & Behavior

| Name | Type | Special Behavior |
|---|---|---|
| ball | circle | Standard win object. |
| tennisball | circle | Light and bouncy. |
| bowlingball | circle | Heavy, low bounce. |
| ballon | circle | Floats upward and pops at height. |
| log | circle | Static, behaves like a round barrier. |
| platform | rectangle | Simple static platform. |
| Domino | rectangle | Dynamic, topples easily. |
| box1 | rectangle | Dynamic box. |
| bucket | bucket | Special static shape. |
| winZone | rectangle | Triggers win if touched by a winning object. |
| winPlat | rectangle | Same as above, but used like a win-sensitive platform. |
| noPlaceZone | rectangle | Building is disabled inside this zone. Objects can still enter during simulation. |

---

### 📥 Example Inventory Snippet

```
{
  "name": "ballon",
  "type": "circle",
  "count": 15,
  "angle": 0.0,
  "size": {
    "radius": 50.0
  },
  "sprite": "madBalloon.png",
  "colour": "#800080",
  "physics": {
    "density": 0.15,
    "friction": 0.9,
    "restitution": 0.2,
    "shape": "DYNAMIC"
  }
}
```

# PUZZLE MODE

In Puzzle Mode, you can play three handcrafted levels created by our Puzzle Masters. We recommend starting with the first level, but you're also free to jump right into any of the others.

You can access the Puzzle Mode from the Title Screen by clicking the *Puzzle* button. This will take you to the *Puzzle Select* screen, where you'll find small preview images on each level card, showing the name and difficulty of the puzzle.

Once you complete a level, you can either return to the Title Screen, export your solution, or jump straight into the next puzzle. After beating the final level (Level 3), you've successfully completed the Puzzle Mode!

# MAD BALLS

Single player

Platform: PC (Linux, Windows, Mac) – Languages: English
Developer: Mad Studios – Publisher: Berger Entertainment
Release Year: 2025